

Durnesswick Laboratories Inc

NM-12 Oral Vehicle: Precipitation-on-Dilution Follow-up

Final Report

Work package DL_2020_0050

Client: Nimbravelle Medicines Inc

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Prepared by: Dr J. Carver, Formulation Development Lead

1. Summary

Nimbravelle needed a 10 mg/mL oral vehicle for NM-12 for repeat-dose rodent studies. The cosolvent-rich vehicle selected earlier gave a clear solution but precipitated within minutes when diluted into simulated gastric fluid. We recommend a dispersed suspension in 0.5% HPMC / 0.1% Tween 80, which tolerated dilution and remained dosable for 24 hours.

2. Vehicle ladder

Eight vehicles were screened at the 10 mg/mL target. Solubility, appearance and 24-hour physical stability are below; this is an exploratory screen, not a locked process.

Table 1 Vehicle ladder results

Vehicle	Solubility (mg/mL)	Appearance	24 h physical stability
0.5% methylcellulose	0.31	Clear, slight yellow tint	Stable
0.5% HPMC / 0.1% Tween 80	1.2	Precipitate at 24 h	Crystal growth at 24 h
20% HP- β -CD in water	1.7	Clear solution	Crystal growth at 24 h
PEG400/water 30:70	14	Fine suspension	Stable
10% DMSO / 90% corn oil	2.1	Precipitate at 24 h	Crystal growth at 24 h
Labrasol/PEG400 1:1	8.8	Clear, slight yellow tint	Stable
5% NMP / 15% Solutol HS15 / 80% water	1.2	Clear, slight yellow tint	Not assessed
pH 4 citrate buffer	3.2	Clear solution	Not assessed

3. Dilution challenge

Each candidate was diluted 1:10 into simulated gastric fluid at 37 °C and observed for 30 minutes. Cosolvent-rich and DMSO-containing vehicles precipitated; the HPMC/Tween suspension and the cyclodextrin solution

did not change visibly. The cyclodextrin vehicle was not carried forward because of the dose-volume loading it would need at the top dose.

4. Recommended vehicle and handling

- Vehicle: 0.5% HPMC / 0.1% Tween 80 in water
- Prepare by wetting the powder with the Tween fraction, then adding HPMC solution in portions with stirring
- Stir continuously during dosing; use within 24 hours of preparation
- Dose volume 10 mL/kg; check homogeneity at top, middle and bottom before the first dose

5. Limitations

Particle size was not controlled; if exposure variability is high in the first study, a milled suspension should be evaluated. Chemical stability of NM-12 in the vehicle beyond 24 hours was not tested.

Name	Role	Signature	Date
Julia Carver	Formulation Development Lead	<i>Carver</i>	09/28/2020
Miguel Serrano	Senior Formulation Scientist	<i>Serrano</i>	09/28/2020